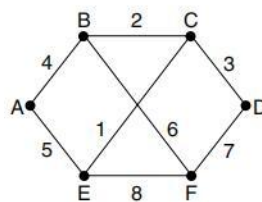


《计算机网络》作业

第八次作业：（英文版教材第五章 5， 7， 14， 15， 16）

5. (5.2.4 Distance Vector Routing) Consider the network of Fig. 5-12(a). Distance vector routing is used, and the following vectors have just come in to router C: from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4). The cost of the links from C to B, D, and E, are 6, 3, and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the cost.



(a)

7. (5.2.6 Hierarchical Routing) For hierarchical routing with 4800 routers, what region and cluster sizes should be chosen to minimize the size of the routing table for a three-layer hierarchy? A good starting place is the hypothesis that a solution with k clusters of k regions of k routers is close to optimal, which means that k is about the cube root of 4800 (around 16). Use trial and error to check out combinations where all three parameters are in the general vicinity of 16.
14. (5.1.3 Implementation of Connectionless Service) A datagram network allows routers to drop packets whenever they need to. The probability of a router discarding a packet is p . Consider the case of a source host connected to the source router, which is connected to the destination router, and then to the destination host. If either of the routers discards a packet, the source host eventually times out and tries again. If both host-router and router-router lines are counted as hops, what is the mean number of
- (a) hops a packet makes per transmission?
 - (b) transmissions a packet makes?
 - (c) hops required per received packet?
15. (5.3.4 Traffic Throttling & 5.3.5 Load Shedding) Describe two major differences between the ECN method and the RED method of congestion avoidance.
16. (5.4.2 Traffic Shaping) Imagine a flow specification that has a maximum packet size of 1000 bytes, a token bucket rate of 10 million bytes/sec, a token bucket size of 1 million bytes, and a maximum transmission rate of 50 million bytes/sec. How long can a burst at maximum speed last?